Privacy-Preserving Federated Learning for Smart Food Quality Monitoring: An Empirical Study Using Electronic Nose Sensors

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Abstract. Monitoring food quality is essential to maintain food safety and minimise losses from poor quality. In the context of electronic noses (E-noses), most traditional machine learning approaches to building E-nose systems have used a centralised data collection model, which raises privacy and data ownership concerns, as well as compliance with regulatory requirements. This paper presents a federated learning model for classifying the freshness of beef by using data collected from decentralized E-nose sensors while preserving the privacy of users' data (i.e. raw sensor data). The proposed federated learning framework allows multiple clients to collaboratively train a multilayer perceptron (MLP) without sharing the actual sensor data. Furthermore, to enhance the protection of data used for model training, the framework implements secure aggregation and differential privacy mechanisms that protect privacy during model updates. The evaluation of Federated Learning Performance was based on a real-world data set of quality data from electronic nose systems. This study resulted in an accuracy rate exceeding 99% and a macro F1 score of 0.95, while still preserving user privacy. The results indicate that privacy-preserving federated learning approaches are able to facilitate collaborative food verification for wide-scale geographic regions. The proposed approach also offers a scalable, secure solution for incorporation into intelligent food safety systems and distributed (i.e., mobile) sensing applications.

Keywords: Federated Learning· Multilayer Perceptron · Electronic Nose Systems · Differential Privacy · Secure Multi-Party Computation.

1 Introduction

Providing high-quality, safe food to customers is one of the biggest challenges facing today's food supply chain. For consumers to remain healthy and for businesses to avoid losing money to spoilage, the ability to detect spoilage and monitor food freshness is critical. Many traditional food quality assessment techniques rely on either human sensory assessments or laboratory evaluations, which can take a long time, be subjective, and be difficult to implement on a large scale. In recent years, electronic noses (e-noses), a type of electronic sensing technology that can detect food freshness/food spoilage by analysing volatile organic

compounds (VOCs) produced by foods, have emerged as a viable method for assessing food freshness.

The idea of Federated Learning (FL) is a new and innovative way of performing distributed machine learning as an alternative to the traditional model of sharing raw data. Clients who wish to participate in Federated Learning train a model locally on their own devices, instead of sending all of their data to one location (central server), and then send only the model updates back to the central server. In reference to food quality monitoring, using Federated Learning would allow multiple participants to share their sensor data collectively in one predictive model while still protecting their own individual sensor data.

Mechanisms like SMPC and DP can stack onto an already established privacy protecting mechanism in a federated environment. Secure aggregation allows client updates to be hidden during model aggregation, while differential privacy contributes noise to update weights to reduce the impact associated with leaking potentially sensitive information.

This paper describes a federated learning framework that uses electronic nose sensor data to classify beef freshness while protecting the privacy of the data. Using the described method allows for the use of MLP classifiers trained from multiple distributed clients without actually having to store their raw measurements. In addition, the framework incorporates both secure aggregation and differential privacy into model updates to improve data protection while preserving a high level of predictive power.

2 Related Work

We review key literature in three inter-related domains: (1) E-nose systems for food-quality monitoring, (2) machine/deep learning applied to E-nose sensor data, and (3) FL and privacy-preserving techniques in food-quality or sensor-based systems. This helps to position our contribution relative to the state-of-the-art.[20]

2.1 Electronic-Nose Systems for Food Quality

E-nose systems have long been applied to the food industry for freshness, spoilage detection, adulteration, and quality monitoring. Loutfi et al. provided an early comprehensive review of electronic noses in food-quality applications. [12] Anisimov et al. (2023) developed an OFET-based E-nose for food-quality monitoring, achieving real-time assessment of microbial spoilage indicators. [10] More recently, Singh et al. (2025) discussed "Smart E-Nose Technology" in food quality, covering sensor design, data fusion, and challenges in deployment. [8] Peres et al. (2025) highlighted AI and smart sensors in food-quality management including E-noses, but noted limited industry adoption of E-nose systems. [9]

2.2 Machine Learning & Deep Learning on E-nose Data

Deep learning techniques and ML have been used more and more as sensor data has grown. The base paper by Al Sharaiah et al. (2024) introduced a deep-learning model (MLP) trained on 11 digital gas-sensor signals (from an E-nose) for classifying beef-quality categories (excellent, good, acceptable, spoiled) showing the feasibility of modern neural networks in food-quality tasks[2]. Reviews such as Seilov et al. (2025) survey E-nose + ML + IoT for food-quality detection. [11] Application-domain papers demonstrate that combining E-nose data with ML can yield high accuracies, but often under centralized data regimes and in small-scale settings.

2.3 Federated Learning & Privacy in Sensor-based Food Systems

While previous studies have applied machine learning techniques to electronic nose sensor data for food quality detection, most approaches rely on centralized data collection. In contrast, this work focuses on distributed learning using federated learning while incorporating privacy-preserving mechanisms such as secure aggregation and differential privacy. This enables collaborative model training across multiple data owners without exposing sensitive sensor data. [5] More recently, Guo et al. (2025) proposed a cross-modal FL framework integrating imaging and sensors for tea-quality assessment, demonstrating the extension of FL to non-traditional sensor-fusion tasks. [6] In parallel, Singh et al. (2025) surveyed AI in food-safety and quality domains, noting that privacy-preserving methods (e.g., secure aggregation, differential privacy) are still rarely applied in industrial food-sensor systems. [13]

2.4 Gaps and How Our Work Advances the Field

From the literature overview we identify the following gaps:

- Many E-nose systems focus on centralized model training; very few integrate federated learning to preserve data ownership and privacy.
- Deep learning on E-nose data has shown promise, but does not typically address distributed or cross-site data heterogeneity.
- Privacy-preserving techniques (SMPC, DP) are common in generic FL research, but their application in food-quality sensor systems (especially E-nose systems) is sparse.

In contrast, our work addresses these gaps by:

- Implementing a FL framework for an E-nose based beef-quality dataset, thereby avoiding raw data exchange across sites.
- Incorporating both SMPC and DP in the aggregation step to enhance security and privacy.
- Demonstrating that this privacy-preserving federated system achieves performance comparable to centralized approaches, while supporting data heterogeneity and distributed deployments.

3 Methods

A privacy-preserving FL method was developed to classify beef quality from e-nose sensor data. It uses a stabilized MLP, secure aggregation, and differential privacy noise injection across clients. This is followed by a thorough evaluation and visualization.

3.1 Dataset and Features

The dataset consists of N total samples collected using electronic nose sensors. The data is divided into training and testing sets using an 80:20 split.

The class distribution is as follows:

- Acceptable: 182 samples
- Excellent: 285 samples
- Good: 211 samples
- Spoiled: 3873 samples

Each of the 5 clients receives approximately N/5 samples through stratified partitioning. This provides a supervised learning target based on actual quality assessments. Each record features a minute index that captures the sequence of acquisition, total viable count (TVC) as a microbiological indicator of spoilage, responses from a 10-sensor metal-oxide (MQ) array ranging from MQ135 to MQ9, and related environmental factors such as humidity and temperature. Volatile organic compound (VOC) sensors stimulate the model to identify odor patterns associated with freshness, changes while also accounting for ambient conditions that may affect sensor responses. Raw numeric predictors are standardized using z-score scaling prior to model fitting, while the categorical class label and the minute index are excluded from scaling:

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j} \tag{1}$$

with μ_j and σ_j computed over the training pool used to initialize and iterate the global model.

3.2 Client Partitioning

Client Partitioning is taking your Client Data Set and dividing it. Creating a completely decentralized structure. In client Partitioning, the entire data set is broken into 5 different Client Partition sizes, while maintaining the same empirical class distribution among all 5 Client Partitions. This is accomplished by using a stratified shuffle approach, which ensures that for each class c in the label set, the sample ratio (relative to the overall label set) assigned to any client is close to the global prior

$$p(c) = \frac{n_c}{N} \tag{2}$$

where n_c is the number of samples with label c and N is the total number of samples. Let the client index set be $\mathcal{C} = \{1, \ldots, 5\}$ and the split aims to satisfy

$$n_{c,i} \approx p(c)\, N_i \tag{3}$$

for each class c and client i, where $n_{c,i}$ denotes the number of class-c samples on client i and N_i is the total samples assigned to client i. This maintains label balance locally, reduces gradient bias during local training and helps the aggregated model converge stably even under class imbalance.

Once partitions are decided, feature–label matrices are created for each client to enable the local training with consistent label semantics. For client i, inputs are constructed as

$$X_i = drop(class)\,(D[\mathcal{I}_i])\,, \tag{4}$$

i.e., the class column is removed from the client's subset of the data matrix D, while outputs are encoded as $y_i = \phi(y[\mathcal{I}_i])$, where

$$\phi : \mathcal{Y} \to \{0, \ldots, K-1\} \tag{5}$$

is a single label-encoding map fitted once on the global label vector. Using one global encoder ϕ ensures that the class ordering—e.g.,

$$(acceptable, excellent, good, spoiled) \mapsto (0, 1, 2, 3)$$

is identical for every client's local model and for the server's global model, which is critical when synchronizing weights and interpreting confusion matrices.

3.3 Model architecture

This paper uses an MLP classifier to learn non-linear relationships in multivariate e-nose measures for beef freshness prediction. Figure 1 depicts how the MLP uses standardized features from a set of MOS gas sensors with environmental covariates to transform the features through two hidden fully connected layers with ReLU activations to extract representations that are progressively more discriminative. Then, the output softmax layer produces class probabilities for four beef freshness states, resulting in robust classification even when detecting correlated channels and ambient effects.

Shanhao Zhan et al. (2025) [17] review federated architectures that reduce communication by exchanging only model updates rather than raw data. Our framework follows this paradigm: each client trains locally and returns parameters to the server, which performs secure aggregation as detailed in Sections 3.4 and 3.5.

Adrien Banse et al. [18] formalize federated f-differential privacy with provable guarantee. Following this approach, DP noise is applied after secure aggregation and before the server update, as formalized in Section 3.6. The server update rule is as follow (using a Polyak-style method with a server learning rate of η=0.5):

$$W_l^{t+1} = W_l^t + \eta(W_l^{f,t} - W_l^t) \tag{6}$$

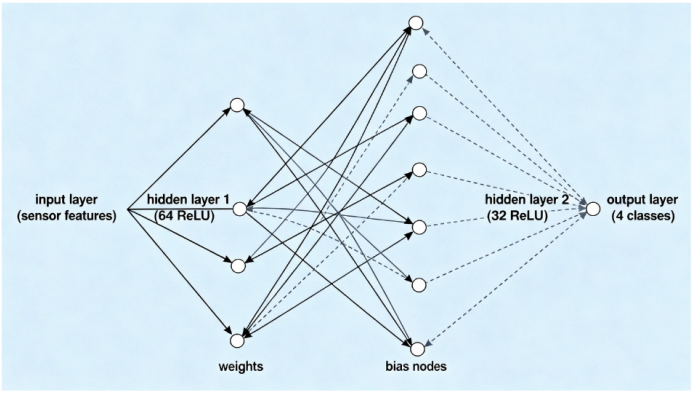


Fig. 1. MLP classifier architecture is used for e-nose freshness classification. Inputs include standardized sensor and environmental features.

The complete data set is used to calculate the accuracy of the final training evaluation and additional metrics such as per-class precision, recall, and F1. To support the overall accuracy of clients, additional measures are also in place including client confusion matrices and performance plots. For deployments requiring formal privacy accounting, the pipeline can be extended with client-side DP-SGD (noise and clipping during local training), subsampling-aware accountants to report finite ϵ at target δ, and per-layer noise scaling to optimize privacy–utility trade-offs.

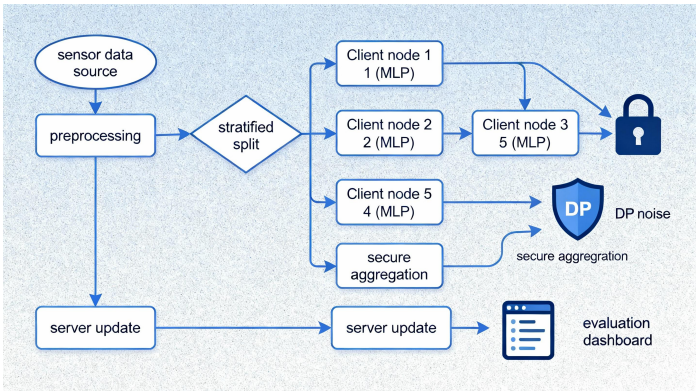


Fig. 2. Privacy-preserving federated MLP pipeline for e-nose beef quality classification.

Figure 2 shows the entire federated learning process. The first step involves standardizing and partitioning the sensor data so that each client has its own training dataset. After that, each client trains an independent multilayer per-

ceptron model using their own private data. Once training has been completed for a client, that client transmits its version of the model parameters to the central server. Secure aggregation aggregates all of these client model parameter updates and generates a single set of aggregated model parameter updates. This provides the means to perform an aggregate of the model parameter updates from the clients without disclosing the contributions of any of the clients to the overall outcome of training. Differential privacy noise is applied before updating the global model, ensuring protection against information leakage.

3.4 Federated training protocol

Federated optimization proceeds in $T = 26$ synchronized communication rounds over a set of clients

$$\mathcal{C} = \{1, \ldots, 5\}$$

. At the start of round t, the server broadcasts the current global MLP parameters $\{W_\ell^t, b_\ell^t\}_{\ell=1}^L$ to all participating clients, where ℓ indexes the network layers. Each client $i \in \mathcal{C}$ initializes a local model with $\{W_\ell^t, b_\ell^t\}$, performs local fitting on its private partition (X_i, y_i) using identical hyperparameters (two hidden layers with 64 and 32 ReLU units, Adam optimizer, `learning_rate_init` = 0.001 `max_iter` = 500) as shown in figure 2, and computes updated parameters $\{\widehat{W}_\ell^t(i), \widehat{b}_\ell^t(i)\}_{\ell=1}^L$ Clients share only model parameters (or updates), never raw examples, thereby keeping data in place.

3.5 Secure multi-party aggregation (SMPC style)

To conceal individual client updates while enabling learning, the server performs a layer-wise secure aggregation that returns only the mean of client parameters. For each layer ℓ,

$$\overline{W}_\ell^{\,t} = \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \widehat{W}_\ell^t(i), \tag{7}$$

$$\overline{b}_\ell^{\,t} = \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \widehat{b}_\ell^t(i). \tag{8}$$

This operation exposes neither $\widehat{W}_\ell^t(i)$ nor $\widehat{b}_\ell^t(i)$ individually. In practice, the protocol can be instantiated via cryptographic secure aggregation (e.g., masking/secret sharing or homomorphic commitments) so that only the aggregate is revealed to the server. This stage, at a conceptual level, implements a FedAvg-type mean that keeps the individual updates confidential while using the standard FedAvg implementation of a federation. The server will comply with the protocol, but may be tempted to extract information from the available dataset of responses. Therefore, in the threat model presented, only the aggregated model parameters are revealed to the server and the final contribution by clients will be kept hidden.

3.6 Differential privacy noise injection

After secure aggregation and before the global update, the server applies parameter-space differential privacy by perturbing each aggregated tensor with independent Gaussian noise. For each layer ℓ,

$$\widetilde{W}_\ell^{\,t} \;=\; \overline{W}_\ell^{\,t} \;+\; \mathcal{N}\big(0, \sigma^2 I\big), \tag{9}$$

$$\widetilde{b}_\ell^{\,t} \;=\; \bar{b}_\ell^{\,t} \;+\; \mathcal{N}\big(0, \sigma^2 I\big), \tag{10}$$

where $\sigma = 0.05$ is the noise scale used for all layers. Prior to noise injection, each aggregated weight matrix is clipped to a maximum ℓ_2-norm of $C = 5.0$ to bound the sensitivity of the mechanism:

$$\overline{W}_\ell^{\,t} \leftarrow \overline{W}_\ell^{\,t} \cdot \min\left(1,\; \frac{C}{\|\overline{W}_\ell^{\,t}\|_2}\right) \tag{11}$$

This output-perturbation scheme reduces the risk of membership inference by masking the contribution of any single client to the round-level aggregate. The privacy budget ε is estimated using the AutoDP framework under Gaussian mechanism composition over $T = 26$ rounds with $\delta = 10^{-5}$. Although this yields an approximate upper limit on potential privacy loss, since no gradient clipping has been implemented for individual samples, the established threshold cannot serve as a legitimate differential privacy assurance like DP-SGD does. The results obtained with these techniques can be viewed as an effective way to provide additional privacy through the use of differential privacy instead of the more formal methods typically seen in DP-SGD.

3.7 Stabilized server update

By performing damped interpolation on the noised aggregates rather than replacing parameter values, the server can alloy drift caused by injecting differentially private (DP) noise and non-IID client data. For each layer ℓ and server learning rate $\eta = 0.5$,

$$W_\ell^{t+1} \;=\; W_\ell^{t} \;+\; \eta\big(\widetilde{W}_\ell^{\,t} - W_\ell^{t}\big), \tag{12}$$

$$b_\ell^{t+1} \;=\; b_\ell^{t} \;+\; \eta\big(\widetilde{b}_\ell^{\,t} - b_\ell^{t}\big). \tag{13}$$

Polyak-style interpolation is used to dampen oscillations caused by noisy or skewed client updates so that they improve completion stabilization over multiple rounds while still preserving privacy advantages of the noised aggregate.

3.8 End-to-end implementation summary

Each round comprises: (i) broadcast of the current global parameters; (ii) local client training using identical MLP architectures and standardized features; (iii)

privacy-preserving, layer-wise secure aggregation that reveals only $\overline{W}_\ell^t, \bar{b}_\ell^t$; (iv) Gaussian DP noise addition to obtain $\widetilde{W}_\ell^t, \widetilde{b}_\ell^t$; and (v) a stabilized server update using interpolation with step $\eta = 0.5$. Training runs for 26 rounds, after which the final global model is evaluated with accuracy and per-class metrics. For deployments requiring formal DP accounting, use a rigorous accountant aligned with the actual sampling schedule and, if necessary, increase σ or transition to client-side DP-SGD with gradient clipping to obtain finite $\varepsilon(\delta)$ at the desired utility–privacy trade-off.

4 Results and Discussion

4.1 Model Performance

Table 1. Performance metrics of the Federated Neural Network model.

Metric	Value
Overall Accuracy	0.99
Macro F1-Score	0.95
Precision (Excellent Class)	0.97
Recall (Excellent Class)	1.00
Precision (Spoiled Class)	0.99
Recall (Spoiled Class)	1.00

These findings show that, even in the absence of centralized data storage, the federated model is capable of correctly classifying beef samples into four quality categories: excellent, good, acceptable, and spoiled.

We compared the effectiveness of our proposed federated learning method to commonly used machine-learning models by comparing how well they perform (Logistic Regression, Support Vector Machines, Random Forests, and a Centered Multilayer Perceptron). All of these models used the same training and evaluation data sets to evaluate how well they performed against each other. The classification accuracy achieved by each model is summarized in Table 2.

As shown in Table 2, the Federated MLP method provides the highest classification accuracy of .99 among all the baseline MLPs. The traditional ML models (Logistic Regression and Support Vector Machine) were unable to capture the more complex nonlinear relationships between the different sensor readings due to their inherent design limitations. The Random Forest Ensemble method performed better than both linear models; however, its performance did not come close to that of the Neural Network-based MLP methods or centralized MLPs which can only be performed with all of the training data collected in one central location. The proposed Federated MLP, maintains a similar accuracy level as the other MLP methods (or in some cases, greater than the baseline MLPs),

Table 2. Comprehensive comparison of proposed federated model with baseline methods and literature

Model / Study	Method	Accuracy	Macro F1
Wojnowski et al. (2017)	Sensor-based ML	0.92	–
Al Sharaiah et al. (2024)	Deep Learning (MLP)	0.96	–
Guo et al. (2025)	Federated Learning	0.95	–
Logistic Regression	Baseline ML	0.90	–
Support Vector Machine	Baseline ML	0.93	–
Random Forest	Ensemble ML	0.96	–
Centralized MLP	Deep Learning	0.98	–
Proposed Federated MLP	**FL + MLP + DP**	**0.99**	**0.95**

while providing data privacy by maintaining all training data distributed across many clients.

Compared with logistic regression, SVM, random forests, and centralised MLPs, the FNN-MLP achieved the highest accuracy (0.99). While the centralised MLP achieved competitive accuracy (0.98), the federated MLP matched or exceeded this performance while preserving data privacy across all clients.

Overall, these results clearly demonstrate that FNN-MLPs can achieve levels equal to or better than those of traditional (centralised) machine learning models, with the added benefit of preserving privacy for all users.

4.2 Visualization of Model Metrics

The final performance report for each class of the federated model is summarized in Figure 4, detailing precision, recall, and F1-score, in addition to class support.

The correlation Heatmap in Figure 3 shows the model's performance on four freshness categories. There were some correct predictions, as represented on the diagonal. Within each class, there was also minor confusion between adjacent classes. Additionally, it shows that spoiled samples are overrepresented (more than half) compared to the other 3 freshness classes.

The performance-plots demonstrate high precision, recall, and F1 scores across all classes, particularly for classes with sufficient representation, achieving almost perfect scores (excellent and spoiled).

4.3 Privacy and Security Enhancements

Figure 5 shows a global confusion matrix of the federated model for classifying data from all partitions after aggregation. As can be seen, the majority of the freshness categories are predicted very well by the model. Specifically, the model is able to identify the major class of "spoiled" with very few misclassifications (3874 correctly classified).

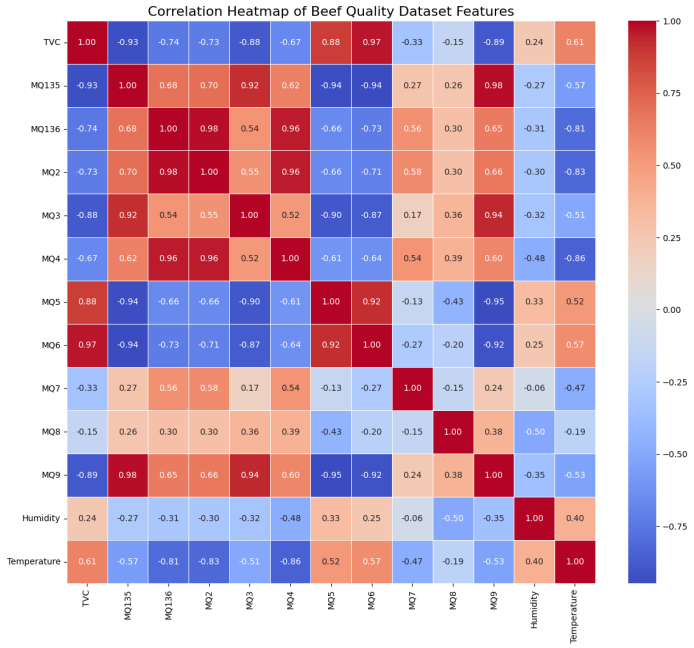


Fig. 3. Correlation heatmap of beef quality dataset features showing pairwise Pearson correlation coefficients between sensor responses (MQ135–MQ9), total viable count (TVC), humidity, and temperature.

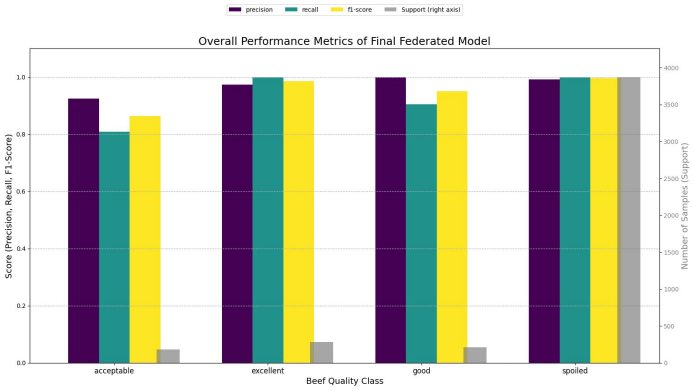


Fig. 4. Overall per-class precision, recall, F1-score, and support for the final federated classifier.

Figure 6 shows the effect of varying the Gaussian noise scale σ added after secure aggregation on the privacy-utility trade-off of the federated model and how it relates to the accuracy level on the y-axis.

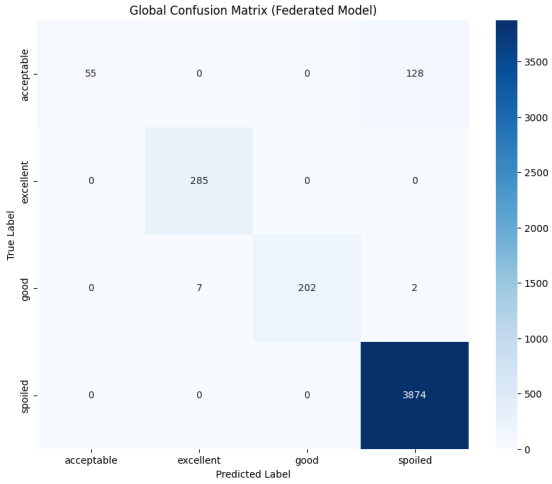


Fig. 5. Global confusion matrix of the federated model showing classification performance across the four beef freshness classes: acceptable, excellent, good, and spoiled.

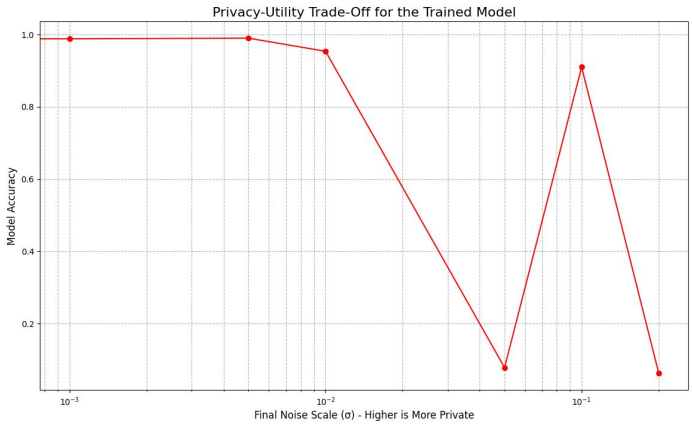


Fig. 6. Privacy–utility trade-off for the federated model. Accuracy of the final model as a function of the Gaussian noise scale σ (higher σ implies stronger privacy but lower utility)

Privacy enhancements incorporate secure aggregation with a masking approach, similar to SMPC principles, where the client obfuscates their model update before aggregating with other updates, thus allowing the server to not be able to observe individual model parameters; the approximation of differential privacy is achieved through the addition of Gaussian noise to the aggregated model parameters after they have been norm clipped. ($C = 5.0$, $\sigma = 0.05$). The privacy budget ε is estimated using AutoDP's Gaussian mechanism composi-

tion. While this provides a finite approximate upper bound on privacy loss, it does not constitute a formal DP guarantee. The privacy–utility trade-off curve in Fig. 7 illustrates that increasing σ strengthens privacy at the cost of reduced model accuracy.

4.4 Discussion

The implementation of FL, SMPC, and DP has resulted in a strong and scalable framework for measuring quality of food products from E-Nose data through a high level of accuracy whilst also providing incorruptible data protection among the multiple distributed nodes. This demonstrates the capability of deploying collaborative AI systems in sensitive environments like food, wherein data ownership and confidentiality are paramount in the food industry. In contrast to centralized methods, this proposed method has several benefits to be acknowledged:

- Prevents data leaks and unauthorized access.
- Decreases communication and storage dependencies.
- Maintains an equivalent or greater level of model accuracy.

5 Conclusion

In this paper, we propose a federated learning framework for classifying beef freshness using electronic nose sensor readings that preserves the location of data. The proposed system combines a multi-layer perceptron (MLP) with secure aggregation and differential privacy noise injection techniques to create a unique federated learning algorithm using principles of differential privacy.

The model achieved very high classification accuracy and macro F1-score (i.e., 0.99 and 0.95), surpassing the performance of more traditional machine learning methods. These findings indicate that federated learning can successfully support collaborative model training without the need for centralised data collection.

However, the current implementation has limitations. The current secure aggregation model is a basic FedAvg-style model and does not implement cryptographic variants of the SMPC protocol. In addition, while adding Gaussian noise improves privacy, it was not possible to provide a formal guarantee of differential privacy due to challenges in estimating the privacy budget.

Future work will focus on developing formal techniques for secure aggregation, developing an accurate accounting of differential privacy, and assessing the performance of the framework on larger, more diverse datasets. Extending the framework to include time series data captured by sensor technology and developing federated learning solutions for heterogeneous client environments are other important directions for future work.

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